Gripenberg-Like Algorithm for Computing the Lower Spectral Radius

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Outline

1. Introduction to the joint and lower spectral radii
2. Antinorms, Gelfand-type limit and convergence theorem
3. Gripenberg-like Algorithm for the lower spectral radius
4. Illustrative Examples
5. Numerical Applications

Joint spectral characteristics of matrix sets

- The **joint spectral radius** (JSR) of a matrix set $\mathcal{F}$ defines the highest possible rate of growth of the products generated by $\mathcal{F}$.
- The **lower spectral radius** (LSR), on the other hand, defines the lowest possible rate of growth.
- For a single matrix A, both coincide with the spectral radius $\rho(A)$.

Joint spectral characteristics of matrix sets

- The **joint spectral radius** (JSR) of a matrix set $\mathcal{F}$ defines the highest possible rate of growth of the products generated by $\mathcal{F}$.

The JSR is an important quantity in many applications:

- Discrete switched systems [Shorten et al., 2007] ($x_{k+1} = A_j x_k$, $j \in \{1, \ldots, n\}$)
- Convergence analysis of subdivision schemes, refinement equations and wavelets [Daubechies and Lagarias, 1991, Daubechies and Lagarias, 1992, Möller and Reif, 2014].
- Numerical stability analysis of ODEs [Guglielmi and Zennaro, 2001].
- Regularity analysis of subdivision schemes [Guglielmi and Protasov, 2016].
- Problems in coding theory and combinatorics [Jungers, 2009].

Joint spectral characteristics of matrix sets

- The **lower spectral radius** (LSR), on the other hand, defines the lowest possible rate of growth.

The LSR is an important **stability** measure; for example:

- In control theory, it identifies the most stable obtainable trajectories, crucial in terms of the behavior of the controlled system [Jungers and Mason, 2017].
- Controls the lower and upper growths of the Euler partition functions [Euler, 1945, Kruyswijk, 1950], which is relevant to the theory of subdivision schemes and refinement equations [Protasov, 2017].

Joint spectral radius

Let $\mathcal{F} = \{A_1, \ldots, A_m\}$ be a matrix family. Let $\|\cdot\|$ be a sub-multiplicative matrix norm and define the quantity:

$$\hat{\rho}_n(\mathcal{F}) := \max_{A_{i_j} \in \mathcal{F}} \|A_{i_n} \cdots A_{i_1}\|, \qquad n \geq 1.$$

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Definition ([Rota and Strang, 1960])

$$\hat{\rho}(\mathcal{F}) := \limsup_{n \to +\infty} \hat{\rho}_n(\mathcal{F})^{1/n} = \inf_{n \geq 1} \hat{\rho}_n(\mathcal{F})^{1/n}$$

is called the **joint spectral radius** (JSR) of $\mathcal{F}$.

Generalized spectral radius

Similarly define a generalization of the quantity $\rho(A^n) = \rho(A)^n$ to the case of several matrices:

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is called the **generalized spectral radius** (GSR) of $\mathcal{F}$.

Berger–Wang formula

Theorem ([Berger and Wang, 1992])

If the matrix family $\mathcal{F}$ is bounded, then GSR and JSR coincide:

$$\hat{\rho}(\mathcal{F}) = \bar{\rho}(\mathcal{F}).$$

Basis for most computational methods used to evaluate JSR/GSR by providing the following lower and upper bounds:

Corollary

For every $n \geq 1$, it turns out that:

$$\bar{\rho}_n(\mathcal{F})^{1/n} \leq \bar{\rho}(\mathcal{F}) = \hat{\rho}(\mathcal{F}) \leq \hat{\rho}_n(\mathcal{F})^{1/n}. \tag{1}$$

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Lower spectral radius via norms

Let again $\|\cdot\|$ be a sub-multilicative matrix norm. Define

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Lower spectral radius via spectral radii

The LSR is independent on the norm, so ([Bochi and Morris, 2015]):

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Since $\rho(A) \leq |A|$ for all induced matrix norms, it follows

$$\check{\rho}(\mathcal{F}) \leq \min_{A_{i_j}\in\mathcal{F}} \rho(A_{i_n}\cdots A_{i_1})^{1/n} \leq \check{\rho}_n(\mathcal{F}), \qquad n \geq 1.$$

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Definition

A product $P := A_{i_n} \cdot A_{i_1}$ is a *spectral lowest product* (s.l.p.) if

$$\check{\rho}(\mathcal{F}) = \rho(P)^{1/n}.$$

Comparison of JSR/LSR properties

- $\hat{\rho}(\mathcal{F}) < 1 \implies$ **stability** of $\mathcal{F}$; $\check{\rho}(\mathcal{F}) < 1 \implies$ **stabilizability** of $\mathcal{F}$.
- The LSR has less stable continuity properties compared to the JSR [Bousch and Mairesse, 2002].
- A direct lower bound analogous to (1) is generally not available for the LSR. However, [Guglielmi and Zennaro, 2020] proved that the LSR can be obtained as a Gelfand-type limit under specific conditions.
- Until recently, "good" properties of the LSR were mainly established for matrix families with invariant cones [Protasov et al., 2010, Jungers, 2012, Guglielmi and Protasov, 2013].
- [Bochi and Morris, 2015] began a systematic investigation of LSR continuity, providing a sufficient condition for **Lipschitz-continuity**.

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  where $A^{\otimes k} := A \otimes \cdots \otimes A$: [Blondel and Nesterov, 2005].
- Sum of squares (**SoS**) techniques for JSR approximation: [Nesterov, 2000, Parrilo, 2000, Parrilo and Jadbabaie, 2007, Legat et al., 2016].

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- Polygonal approximation of extremal norms: [Protasov, 2016], [Guglielmi and Zennaro, 2005], [Guglielmi and Protasov, 2013].

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  - Construct a special polytope antinorm starting from P.
- Convergence is achieved if P is a s.l.p.; otherwise, the algorithm restarts from a product of length $n' > n$.

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This approach does not require identifying a s.l.p. or construct special antinorms; instead:

- guarantees convergence from any initial (polytope) antinorm;
- aims to use the smallest possible set of products to estimate the LSR, thus reducing the computational cost.

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- guarantees convergence from any initial (polytope) antinorm;
- aims to use the smallest possible set of products to estimate the LSR, thus reducing the computational cost.

We also introduce an **adaptive procedure** to refine the initial antinorm iteratively, significantly improving speed/accuracy.

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Invariant cones and asymptotically rank-one matrices

A proper cone (positively homogeneous, salient and solid) K is:

- **invariant** for $\mathcal{F}$ if $A(K) \subseteq K$ for all $A \in \mathcal{F}$;
- **strictly invariant** for $\mathcal{F}$ if $A(K \setminus \{0\}) \subseteq \text{int}(K)$ for all $A \in \mathcal{F}$.

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Let $\mathcal{F}$ be a family of real-valued matrices. Define the set of all products

$$\Sigma(\mathcal{F}) := \bigcup_{k \geq 1} \Sigma_k(\mathcal{F}), \qquad \Sigma_k(\mathcal{F}) := \left\{ \prod_{j=1}^{k} A_{i_j} \ : \ A_{i_j} \in \mathcal{F} \right\}.$$

Definition

A family $\mathcal{F}$ is **asymptotically rank-one** if all products $P \in \Sigma(\mathcal{F})$ have:

- $\rho(P) > 0$ and either $\rho(P)$ or $-\rho(P)$ is a simple eigenvalue;
- $|\lambda| < \rho(P)$ strictly for every $\lambda \in \Lambda(P)$.

Polytope antinorms

An antinorm is a continuous function $a(\cdot)$ defined on a cone K satisfying:

- $a(x) \geq 0$ for all $x \in K$;
- $a(\lambda x) = \lambda a(x)$ for all $\lambda \geq 0$ and all $x \in K$;
- $a(x + y) \geq a(x) + a(y)$ for every $x, y \in K$.

The **unit antiball** of $a(\cdot)$ is $\mathcal{A} := \{x \in K \ : \ a(x) \geq 1\}$.

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Definition (Polytope Antinorm)

$a(\cdot)$ is a polytope antinorm if $\mathcal{A}$ is a positive infinite polytope. Specifically, there is a minimal **vertex set** $V = \{v_1, \ldots, v_p\} \subset K \setminus \{0\}$ such that:

- $a(v_i) = 1$ for every $i = 1, \ldots, p$;
- $\mathcal{A} = \text{convhull}(V) + K$.

Gelfand's limit (part one)

Let $\mathcal{L}(K)$ be the set of all $\mathbb{R}^{d,d}$-matrices for which K is invariant and $a(\cdot)$ an antinorm. For $A \in \mathcal{L}(K)$, we define the induced operator antinorm:

$$a(A) := \inf\{a(Ax) \ : \ x \in K \text{ s.t. } a(x) = 1\}.$$

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Definition (Antinorm of matrix set)

The antinorm of $\mathcal{F} = \{A_1, \ldots, A_m\}$ is

$$a(\mathcal{F}) := \min_{1 \leq i \leq m} a(A_i).$$

Lemma (Lower bound for the LSR)

$$a(\mathcal{F}) \leq \check{\rho}(\mathcal{F}).$$

Gelfand's limit (part two)

Theorem ([Guglielmi and Protasov, 2013])

Let $\mathcal{F}$ and K be as above. Then

$$\check{\rho}(\mathcal{F}) = \max\{a(\mathcal{F}) \ : \ a(\cdot) \text{ antinorm defined on } K\}.$$

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Strategy: Fix an antinorm $a(\cdot)$ and, for $k \geq 1$, we define

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Theorem ([Guglielmi and Zennaro, 2020])

For all $k \geq 1$, it turns out that $\alpha_k(\mathcal{F}) \leq \check{\rho}(\mathcal{F})$. In addition, if either $\check{\rho}(\mathcal{F}) = 0$ or $\alpha_1(\mathcal{F}) > 0$, there holds

$$\alpha(\mathcal{F}) := \lim_{k \to +\infty} \alpha_k(\mathcal{F}) = \sup_{k \geq 1} \alpha_k(\mathcal{F}). \tag{2}$$

Gelfand's limit (part three)

In general, the inequality $\alpha(\mathcal{F}) \leq \check{\rho}(\mathcal{F})$ may be **strict**. Thus, we assume that $\mathcal{F}$ is asymptotically rank-one and associate the unique decomposition

$$\mathbb{R}^d = V_P \oplus H_P$$

to every $P \in \Sigma(\mathcal{F})$. Define the *leading set* $\mathcal{V}(\mathcal{F}) = \cup_{P \in \Sigma(\mathcal{F})} V_P$, where V_P is the leading eigenspace of P, and the *secondary set*

$$\mathcal{H}(\mathcal{F}) = \bigcup_{P \in \Sigma(\mathcal{F})} H_P.$$

Gelfand's limit (main result)

Theorem (Gelfand's limit, [Guglielmi and Zennaro, 2020])

Let $\mathcal{F}$ be an asymptotically rank-one family and assume that

$$K \cap \operatorname{clos} \mathcal{H}(\mathcal{F}) = \{0\}. \tag{3}$$

Then, for any positive antinorm a, there holds:

$$\alpha(\mathcal{F}) := \lim_{k \to +\infty} \alpha_k(\mathcal{F}) = \check{\rho}(\mathcal{F}). \tag{4}$$

In particular, if $\mathcal{F} = \{A\}$, then

$$\rho(A) = \lim_{k \to +\infty} a(A^k)^{1/k}.$$

Continuity properties

The continuity properties of the LSR are not completely understood. For instance, discontinuities may occur outside of invariant cones:

Example ([Bochi and Morris, 2015])

Let $\mathcal{F} = \{A_1, A_2\}$ be the family given by the matrices

$$A_1 = \begin{pmatrix} 2 & 0 \\ 0 & 1/8 \end{pmatrix} \qquad \text{and} \qquad A_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

satisfying $\check{\rho}(\mathcal{F}) = 1$. For $N \in \mathbb{N}$, let $\mathcal{F}_N = \{A_1, R_{2\pi/N}\}$, where $R_{2\pi/N}$ is the rotation matrix of angle $2\pi/N$. For $m \geq 1$, we have

$$(A_1)^m (R_{2\pi/N})^N = \begin{pmatrix} 2^m & 0 \\ 0 & 8^{-m} \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -2^m \\ 8^{-m} & 0 \end{pmatrix},$$

which implies $\check{\rho}(\mathcal{F}_N) = 1/2$, proving the discontinuity of $\check{\rho}$ at $\mathcal{F}$.

Continuity result

Definition ([Protasov et al., 2010])

A convex closed cone K' is embedded in a cone K if $K' \setminus \{0\} \subset \operatorname{int} K$. If they both are invariant for $\mathcal{F}$, then $\{K, K'\}$ is an **invariant pair** for $\mathcal{F}$.

Theorem ([Jungers, 2012])

Let $\mathcal{F} \subset \mathbb{R}^{d,d}$ be compact and let $\mathcal{F}_\epsilon$ be a sequence that converges to $\mathcal{F}$ in the Hausdorff metric. If $\mathcal{F}$ admits an invariant pair of cones, then

$$\check{\rho}(\mathcal{F}_\epsilon) \xrightarrow{\epsilon \to 0} \check{\rho}(\mathcal{F}). \tag{5}$$

Corollary

If $A > 0$ for every $A \in \mathcal{F}$, then (5) holds.

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Algorithm **(S)**: setting

Fix $\delta > 0$. Define $S_1 := \Sigma_1 = \mathcal{F}$ and set the initial lower/upper bounds:

$$t_1 := \min_{A \in S_1} a(A) \leq \min_{A \in S_1} \rho(A) =: s_1.$$

For $k > 1$, consider the following subset of $\Sigma_k(\mathcal{F})$

$$S_k := \left\{ P = \prod_{\ell=1}^{k} A_{i_\ell},\ i_\ell \in \{1, \dots, m\} \quad \text{s.t.} \quad \prod_{\ell=1}^{k-1} A_{i_\ell} \in S_{k-1} \text{ and } q(P) < s_{k-1} - \delta \right\},$$

where $q(P) := \max_{1 \leq j \leq k} a\left(\prod_{\ell=1}^{j} A_{i_\ell}\right)^{1/j}$, and

$$s_k := \min\left\{ s_{k-1}, \min_{P \in S_k} \rho(P)^{1/k} \right\}, \quad t_k := \max\left\{ t_{k-1}, \min\left\{ s_k - \delta, \min_{P \in S_k} q(P) \right\} \right\}.$$

Convergence theorem

Theorem

Let $\mathcal{F}$ be an asymptotically rank-one family sharing an invariant cone K satisfying (3) and fix $\delta > 0$. Then, it turns out that

$$t_k \leq \check{\rho}(\mathcal{F}) \leq s_k \qquad \text{for every } k \in \mathbb{N}.$$

Additionally, the sequences of differences converges to the fixed accuracy:

$$\lim_{k \to +\infty} (s_k - t_k) = \delta.$$

Adaptive variant: Algorithm **(A)**

- Let $a^{(0)} := a$ be the initial polytope antinorm, and let V_0 be the corresponding minimal vertex set.
- Suppose that $a^{(k-1)}$ and V_{k-1} are known and set $V_k := V_{k-1}$. Start the next step of the algorithm and, for every $P \in S_k$, let $y \in \mathbb{R}^d$

$$\min_{\{x\,:\,a^{(k-1)}(x)=1\}} a^{(k-1)}(Px) = a^{(k-1)}(Py)$$

  Add the vector $z := Py$ to V_k if it lies inside the current polytope and discard it otherwise.
- Extract from V_k a minimal vertex set (**pruning procedure**) and let $a^{(k)}(\cdot)$ be the corresponding polytope antinorm.

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Drawing inspiration from the **polytopic algorithm**:

(i) At step k, identify all products $P \in S_k^*$ that yield an improvement in the upper bound.

(ii) Fix $\vartheta > 1$ and let v_P be the leading eigenvector of P. Compute $a(v_P)$ and add the rescaling $w_P := v_P/(\vartheta \cdot a(v_P))$ to the vertex set.

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Remark

- *Including eigenvectors in the vertex set potentially improves convergence, but it is* **not** *guaranteed.*
- *The choice of the scaling parameter ϑ plays an important role, but there is no theoretical result on how to choose it.*

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  - ℓ_{slp}: the degree that optimizes the upper bound only;
  - $\mathtt{n}_{op}$: the number of antinorm evaluations performed;
- The antinorm of a matrix is computed by solving LP optimization problems. Thus, adaptive algorithms include a pruning procedure to keep $\#V$ as small as possible.

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Illustrative example

In [Brundu and Zennaro, 2018], they show that the family

$$\mathcal{F} := \{A_1, A_2\} = \left\{ \begin{pmatrix} 7 & 0 \\ 2 & 3 \end{pmatrix}, \begin{pmatrix} 2 & 4 \\ 0 & 8 \end{pmatrix} \right\}$$

has a s.l.p. given by $\Pi := A_1 A_2 (A_1^2 A_2)^2$ and, thus, LSR:

$$\check{\rho}(\mathcal{F}) = \rho(\Pi)^{1/8} = \left[4 \cdot \left(213803 + \sqrt{(44666192953)} \right) \right]^{1/8} \approx 6.009313489.$$

Thus, we consider the rescaled family:

$$\widetilde{\mathcal{F}} := \rho(\Pi)^{-1/8} \cdot \mathcal{F} \quad \text{and} \quad \widetilde{\Pi} := \rho(\Pi)^{-1} \cdot \Pi.$$

Algorithm **(S)**

The convergence is slow: the lower bound improves from $M = 50$ to $M = 10^6$ rather slowly, increasing by only approximately 1.4%.

M	**low.bound**	**up.bound**	ℓ_{slp}	ℓ_{opt}
5×10^1	0.985 087	1.000 025	5	7
1×10^3	0.995 985	1.000 000	8	17
1×10^4	0.997 680	1.000 000	8	27
1×10^6	0.998 927	1.000 000	8	51

Algorithm **(A)**

Even taking $M = 50$ leads to convergence to an accuracy that is smaller than the ϵ-precision of the machine:

$$\texttt{lower bound} = \texttt{upper bound} = 1.$$

It identifies the correct s.l.p. degree (8) and returns three s.l.p. candidates:

$$\begin{pmatrix} 0.5690 & 0.6204 \\ 0.3912 & 0.4367 \end{pmatrix}, \quad \begin{pmatrix} 0.6376 & 0.5068 \\ 0.4519 & 0.3682 \end{pmatrix}, \quad \begin{pmatrix} 0.7667 & 0.9168 \\ 0.1936 & 0.2391 \end{pmatrix}.$$

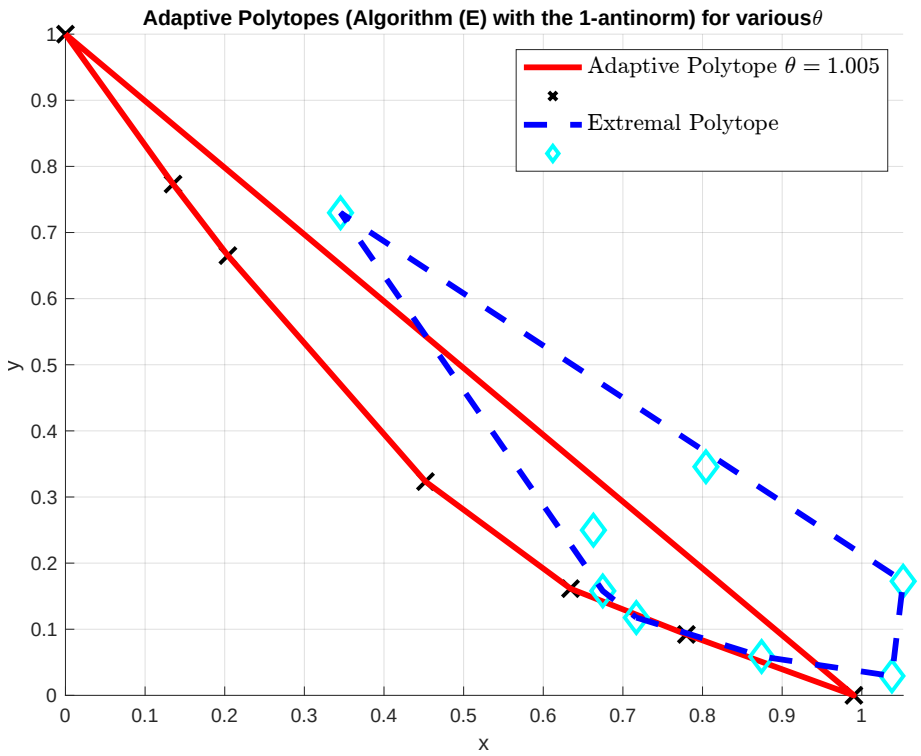
which are, respectively, $\widetilde{\Pi}$ and two of its cyclic permutations.

Algorithm **(E)**

Convergence similar to Algorithm **(A)**, so we focus only on ϑ:

ϑ	**low.bound**	**up.bound**	$(\ell_{slp}, \ell_{opt}, \mathtt{n}, \mathtt{n}_{op}, \mathtt{J}_{max})$	$\#V$
1.005	1	1	(8,10,10,50,5)	7
1.205	1	1	(8,18,18,174,9)	20
1.305	1	1	(8,22,22,380,8)	57
1.405	1	1	(8,24,24,392,24)	58
1.605	1	1	(8,20,20,362,24)	53
1.805	1	1	(8,19,19,334,24)	46

- The algorithm converges for all values of ϑ tested.
- Small scaling parameters (i.e., $\vartheta = 1.005$) are better: low number of evaluations ($\mathtt{n}_{op} = 50$) and return an antinorm with 7 vertices close to the extremal one.



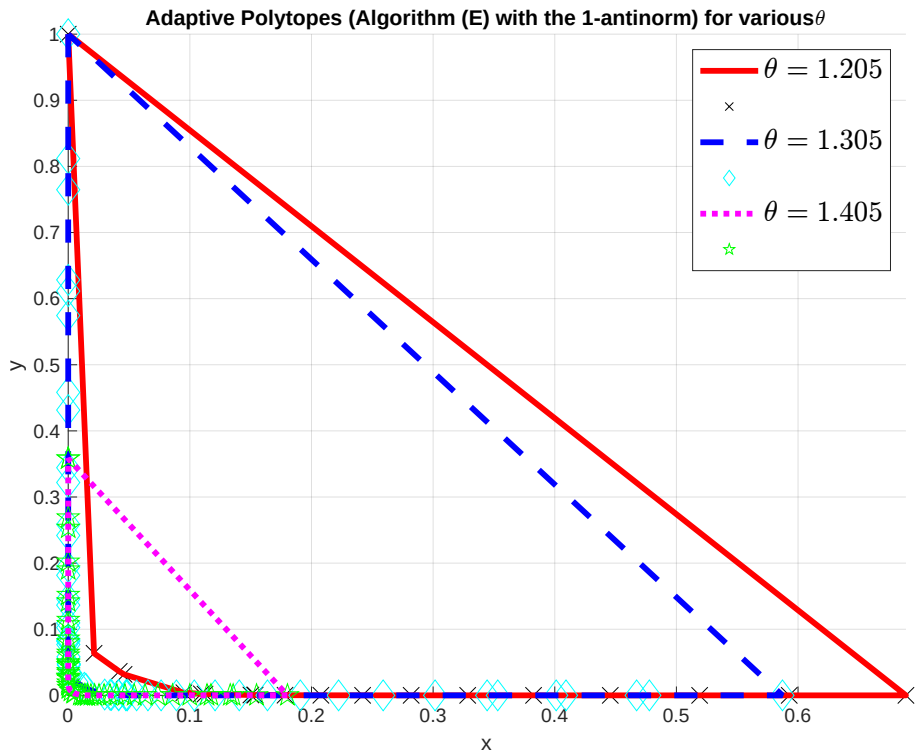


Figure: Comparison between extremal and adaptive ($\vartheta = 1.005$) polytopes (**left**). Comparison of adaptive polytopes generated for different values of ϑ (**right**).

Critical case

The family given by

$$A_1 = \begin{pmatrix} 5 & 1 & 0 & 0 \\ 0 & 5 & 2 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix} \quad \text{and} \quad A_2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 5 & 6 \\ 0 & 0 & 3 & 7 \\ 0 & 0 & 0 & 4 \end{pmatrix}$$

is a critical example: A_1 has a non-simple leading eigenvalue ($\lambda = 5$) and secondary eigenvectors lie on the boundary of $\mathbb{R}^d_+$.

- The Gelfand's limit is not satisfied with an arbitrary antinorm. For example, the 1-antinorm has $a_1(A_2^k) = 1$ for all $k \in \mathbb{N}$ so

$$\lim_k \alpha_k = 1 \neq 3 = \check{\rho}(\mathcal{F}).$$

Proposal: use the polytope antinorm corresponding to $V = \{v_1\}$, where v_1 is the leading eigenvector of A_1 or A_2.

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Numerical results

Consider the rescaled family $\tilde{\mathcal{F}} := \mathcal{F}/3$. Then:

- Algorithm **(S)** converges, but too slowly: lower bound of 0.889231 and an upper bound of 1 with 10^6 evaluations.
- The adaptive Algorithm **(A)** converges to the selected accuracy $\delta = 10^{-6}$ within 512 antinorm evaluations. However, the vertex set grows to 1020 vertices despite the pruning procedure.
- Algorithm **(E)** fails to return a lower bound because vertices of the form $(\star, 0, 0, 0)^T$ appear, causing the algorithm to stop prematurely.

Perturbation theory

Regularize the family using small perturbations of the form:

$$\tilde{\mathcal{F}}_\epsilon^\Delta = \tilde{\mathcal{F}} + \epsilon\{\Delta_1, \Delta_2\},$$

where $\epsilon > 0$ and Δ_1, Δ_2 are non-negative matrices with $\|\Delta_i\|_F = 1$.

ϵ	Algorithm **(A)**		Algorithm **(E)**	
	low.bound	**up.bound**	**low.bound**	**up.bound**
1×10^{-3}	1.001 520	1.109 109	1.028 864	1.109 109
1×10^{-4}	0.999 286	1.046 703	0.999 286	1.046 703
1×10^{-5}	0.997 308	1.018 235	0.997 517	1.018 235
1×10^{-6}	0.994 291	1.005 660	0.994 587	1.005 660
1×10^{-7}	0.998 546	1.001 086	0.999 128	1.001 086

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Euler binary partition

Let $r \geq 2$ integer. The Euler partition function $b(k) := b(r, k)$ is defined as the number of distinct binary representations:

$$k = \sum_{j=0}^{+\infty} d_j 2^j, \qquad \text{where } d_j \in \{0, \ldots, r-1\} \text{ for all } j \in \mathbb{N}.$$

The function's asymptotic behavior is characterized in [Reznick, 1990], for r odd, via the JSR ($\hat{\rho}$) and LSR ($\check{\rho}$) of a matrix family $\mathcal{F}$ as follows:

$$\limsup_{k \to +\infty} \frac{\log b(k)}{\log k} = \log_2 \hat{\rho}(\mathcal{F}) \qquad \text{and} \qquad \liminf_{k \to +\infty} \frac{\log b(k)}{\log k} = \log_2 \check{\rho}(\mathcal{F}).$$

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The family $\mathcal{F}$ is given by two $(r-1) \times (r-1)$ matrices which, for $s \in \{1, 2\}$, are defined by the following relation:

$$(A_s)_{ij} = \begin{cases} 1 & \text{if } i + 2 - s \leq 2j \leq i + r - s + 1, \\ 0 & \text{otherwise.} \end{cases}$$

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For instance, for $r = 7$ the family $\mathcal{F}$ is given by

$$A_1 = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{pmatrix} \qquad \text{and} \qquad A_2 = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

Some remarks on the problem

- In [Guglielmi and Protasov, 2013], the polytopic algorithm demonstrates that for r odd (up to 41), A_1 or A_1A_2 is a s.l.p.
- The 1-antinorm is not suitable as the cone $\mathbb{R}^d_+$ is not strictly invariant for $\mathcal{F}$ due to the many zero entries.

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To apply the algorithm, we follow a simple strategy:

(a) Apply Algorithm **(A)** with M small (e.g., $M = 10$) to establish a preliminary bound $L \leq \check{\rho}(\mathcal{F}) \leq H$.

(b) Rescale using the preliminary lower bound.

(c) Iteratively apply Algorithm **(A)** starting from the antinorm refined at the previous step. Use the stopping criterion:

$$H_{new} - L_{new} < \delta \qquad \text{or} \qquad |L_{new} - L| < \epsilon. \tag{6}$$

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Numerical results: Algorithm (A) vs Polytopic algorithm

r	low.bound	up.bound	T[s]	$\#V_{adaptive}$	$\#V_{extremal}$
7	3.490 363	3.491 891	78	12	14
9	4.493 916	4.494 493	200	11	17
15	7.499 776	7.499 842	261	23	23
17	8.499 875	8.499 904	260	28	30
19	9.499 646	9.499 789	340	24	46
23	11.499 848	11.499 894	317	33	31
25	12.499 932	12.499 971	393	42	58
29	14.499 962	14.499 982	344	24	43
31	15.499 997	15.499 999	302	23	34
35	17.499 980	17.499 997	310	22	102
37	18.499 963	18.499 994	341	25	113
39	19.499 986	19.499 994	292	18	59
41	20.499 983	20.499 997	336	23	120
251	125.499 991	125.500 000	16	11	
501	250.499 997	250.500 000	20	7	
751	375.500 000	375.500 000	2	17	
3001	1500.500 000	1500.500 000	4	1	

Random sparse matrices

- Randomly generated families with sparsity density $\rho \in \{0.25, 0.5, 0.75\}$.
- The initial antinorm is given by the leading eigenvector of A_1 as the 1-antinorm is not suitable with numerous zero entries.

ρ	d	low.bound	up.bound	$(\ell_{slp}, \ell_{opt}, \mathtt{n}_{op})$	$\#V$
0.25	25	2.531 959	2.531 959	$(3, 7, 254)$	8
	50	5.425 970	5.425 970	$(2, 8, 510)$	29
	100	11.229 93	11.229 93	$(2, 6, 126)$	19
	150	16.338 661	16.338 661	$(2, 8, 510)$	8
	200	22.194 604	22.194 604	$(2, 4, 30)$	8

- For $\rho = 0.25$, the algorithm typically incorporates more vertices compared to higher density. Furthermore, a relative accuracy of $\delta = 10^{-6}$ generally requires a higher number of operations $\mathtt{n}_{op}$.

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	150	16.338 661	16.338 661	$(2, 8, 510)$	8
	200	22.194 604	22.194 604	$(2, 4, 30)$	8

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ρ	**d**	**low.bound**	**up.bound**	$(\ell_{slp}, \ell_{opt}, \mathtt{n}_{op})$	$\#V$
0.50	25	4.700 123	4.700 123	$(4, 6, 126)$	13
	50	9.771 280	9.771 280	$(3, 4, 30)$	4
	100	19.748 468	19.748 468	$(1, 4, 30)$	7
	150	29.331 904	29.331 904	$(2, 7, 254)$	7
	200	39.459 047	39.459 047	$(2, 6, 126)$	12
0.75	25	6.480 977	6.480 977	$(2, 6, 126)$	4
	50	12.811 969	12.811 969	$(1, 5, 62)$	4
	100	26.476 497	26.476 497	$(2, 5, 62)$	4
	150	39.447 342	39.447 342	$(1, 6, 126)$	4
	200	52.924 237	52.924 237	$(1, 4, 30)$	4

- For $\rho \in \{0.5, 0.75\}$, the algorithm exhibits significantly faster convergence and incorporates fewer vertices. Additionally, s.l.p. are always identified correctly.

Sparse families with sparsity density $\rho = 0.1$

d	ϵ	**low.bound**	**up.bound**	**T**[s]	#**vert**	(ℓ_{slp}, ℓ_{opt})
25	1×10^{-3}	0.773 553	0.773 554	311	38	(4, 14)
	1×10^{-5}	0.772 502	0.772 503	439	46	(1, 16)
	1×10^{-7}	0.772 492	0.772 493	454	43	(4, 42)
50	1×10^{-3}	2.223 289	2.223 293	1850	52	(1, 11)
	1×10^{-5}	2.222 668	2.222 702	1518	59	(1, 9)
	1×10^{-7}	2.222 460	2.222 696	1418	52	(1, 9)
100	1×10^{-3}	4.608 453	4.608 458	1489	55	(3, 10)
	1×10^{-5}	4.606 516	4.607 851	2296	47	(3, 9)
	1×10^{-7}	4.607 840	4.607 845	2456	57	(3, 10)
200	1×10^{-3}	9.476 520	9.476 530	165	8	(2, 21)
	1×10^{-5}	9.475 914	9.475 924	137	8	(3, 9)
	1×10^{-7}	9.475 908	9.475 918	148	10	(1, 13)

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